



KENYATTA UNIVERSITY
SCHOOL OF EDUCATION DEPARTMENT
OF CHEMISTRY

BSc Ed., BSc. and BSc. Analytical & Industrial Management

SCH 211 CHEMICAL THERMODYNAMICS

LEVEL : II SEMESTER: I ACADEMIC YEAR: 2020/2021

Series -1 (Introduction and First Law of Thermodynamics)

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Course Objective & Content

Course objective:

To provide a basic introduction to chemical thermodynamics for science and education students.

Course Content:

Chemical thermodynamic systems, states, state function and equilibrium state. First law of thermodynamics, constant volume, constant pressure and the reversible process. Isothermal and adiabatic expansion of an ideal gas. Thermochemistry: Heat changes involved in chemical reactions and Hess's law. Heat capacities and enthalpy dependence on temperature.

Second law of thermodynamics: entropy and disorder. Temperature dependence on entropy and the third law of thermodynamics.

Free energy (G), pressure dependence of G on ideal gases, relationship between G and the equilibrium constant.

Learning objectives:

By the end of this lesson you should be able to:

i) Define:

- a) System, surroundings, universe
- b) Open, closed and isolated systems
- c) Energy, heat and work
- d) Isothermal, adiabatic, reversible and irreversible processes or changes

- e) Thermodynamic functions such as internal energy (E) and enthalpy (H)
- f) Heat capacity at constant volume (C_v), and Heat capacity at constant pressure (C_p) and relate the two. ii) Solve problems relating to thermodynamics, for example:
 - a) The change, during expansion or compression of an ideal gas, of Enthalpy, internal energy and the work done.
 - b) Adiabatic changes in systems

Learning objectives (Cont-1):

- iii. Relate heats of reaction at constant pressure and constant volume
 - a) Calculate the change in enthalpy accompanying a chemical reaction
 - b) Apply the enthalpies of combustion, formation and thermochemical laws in solving problems
- iv. Demonstrate the dependence of enthalpies of reaction on temperature

- v. Define entropy and state the second law of thermodynamics
 - a) Calculate the change of entropy when an ideal gas undergoes an expansion or compression
 - b) Calculate the entropy changes of systems and surroundings during temperature/pressure/volume change
 - c) Calculate the entropy changes during phase transitions
 - d) Calculate the entropy changes for irreversible processes
 - e) Describe the Carnot cycle and calculate efficiency of heat engines
 - f) Relate entropy changes to irreversibility and spontaneity of transformations

Learning objectives (Cont-2):

- vi. Define Helmholtz and Gibb's Free Energies
 - a) Calculate the Gibb's Free Energy changes for different physical and chemical transformations.

- b) Relate the Gibb's Free Energy changes to Spontaneity of transformations.
- c) Relate changes in Gibb's Free Energy with Temperature, Enthalpy, Entropy and Equilibrium Constant.
- d) Apply the Clausius-Clapeyron Equation

Suggested References

- 1) Glasstone S., Lewis D. Elements of Physical Chemistry, Macmillan, 2nd. Edition, 1974.
- 2) Barrow G. M., Physical Chemistry, McGraw-Hill, 3rd Edition, 1973
- 3) Sharma K. K., Sharma L.K. A Textbook of Physical Chemistry, Vika's
- 4) Bahl B.S., Tuli G.D., Essentials of Physical Chemistry, Chand and Co., 1985

- 5) Mahan B.H. College Chemistry, Addison Wesley
- 6) Griffith's P.J.F., Thomas J.D.R., Calculations in Advanced Physical Chemistry, Edward Arnold.
- 7) Ritchie I.M. Graig R.A., Thistlethwaite P.J. Problems in Physical Chemistry J.Wailey
- 8) Atkins P.W., Physical Chemistry, ELBS
- 9) Dickerson R.E., Molecular Thermodynamics, W.A. Benjamin
- 10) Levine I.N. Physical Chemistry

1.0 INTRODUCTION

The term **THERMODYNAMICS** means flow of heat. The study of energy and its transformations.

(Greek: *therm*, 'heat' *dynamis* 'power'). In general it deals with the inter conversion of various kinds of energy in physical and chemical systems.

Thermodynamics

1. Predicts the feasibility of a physical process or chemical reaction under given condition of temperature and pressure
2. Predicts whether a chemical reaction would occur spontaneously or not under a given set of conditions
3. Helps to determine the extent to which a reaction would take place.

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Introduction

Limitations of Thermodynamics

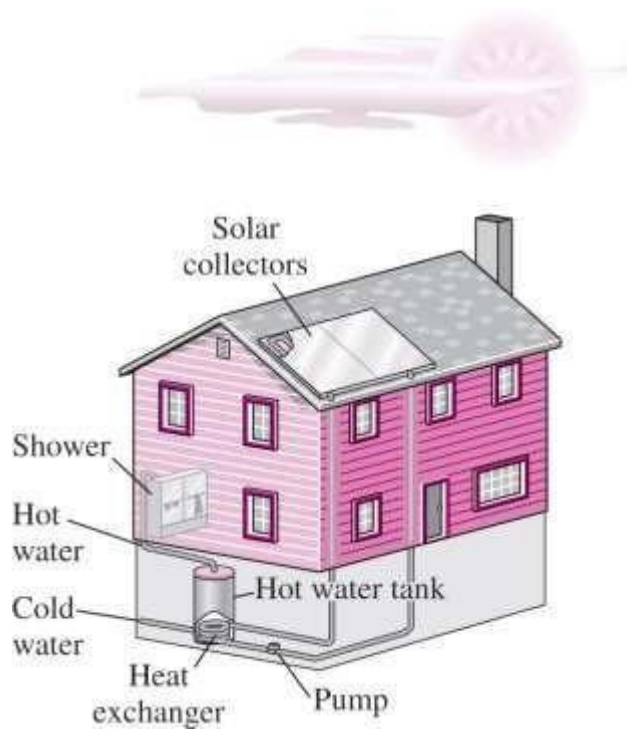
1. It predicts the extent to which a reaction can take place. However it does not say anything about the rate.
2. It applies only to matter in bulk and not to individual atoms or molecules

Importance of thermodynamics

Thermodynamics is important not only to chemistry but to other areas of science and engineering as well. It touches our daily lives as we use energy for manufacturing, travel and communications.

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Pictorial Application Areas of Thermodynamics



1.1 THE NATURE OF ENERGY

Definitions

FORCE

A force is any kind of push or pull exerted on an object. For example force of gravity ‘pull’ us to the ground. In chemistry, positively charged nuclei exert a ‘pull’ on negatively charged electrons.

WORK

When we increase the distance between a proton and an electron, we perform work to overcome the attractive force between them. The work, w , that we do in moving objects against a force equals the product of the force, F , and the distance, d , that the object is moved. $w = F \times d$

1.1 THE NATURE OF ENERGY

Definitions

ENERGY

Energy in the form of work, must be used to move an object against a force. As we perform work, heat may be generated.

HEAT

Heat is the energy that is transferred from one object to another because of difference in temperature.

WORK and HEAT are the two ways that we experience energy changes in our MACROSCOPIC environment. Energy is the capacity to do work or to transfer heat

1.1 THE CONCEPT OF ENERGY

ENERGY may take various forms:

- kinetic energy (due to motion)
- potential energy (due to position)
- heat energy (as measured by temperature)
- chemical energy, etc

A basic definition given by the great physicist, Albert Einstein is:

$$E = mc^2$$

where

E = energy, m = mass and c = velocity of light. **Some**

SI derived Units

Physical Quantity	Unit	Symbol	Definition of unit
force	newton	N	$1\text{N} = 1\text{mkgs}^{-2}$
pressure	pascal	Pa	$1\text{Pa} = 1\text{Nm}^{-2} = 1\text{kgm}^{-1}\text{s}^{-2}$
Celcius temperature	degree Celcius	$^{\circ}\text{C}$	$t/^{\circ}\text{C} = T/\text{K} - 273.15$
energy	joule	J	$1\text{J} = 1\text{NM} = 1\text{m}^2\text{kgs}^{-2}$
power	watt	W	$1\text{W} = 1\text{Js}^{-1} = 1\text{m}^2\text{kgs}^{-3}$

frequency	hertz	Hz	$1\text{Hz} = 1\text{s}^{-1}$
electric charge	coulomb	C	$1\text{C} = 1\text{As}$
electric potential	volt	V	$1\text{V} = 1\text{JC}^{-1} = 1\text{m}^2\text{kgs}^{-3}\text{A}^{-1}$
<u>electric resistance</u>	ohm	Ω	$1\Omega = 1\text{VA}^{-1} = 1\text{m}^2\text{kgs}^{-3}\text{A}^{-2}$

1.2 BASIC DEFINITIONS

System and surroundings:

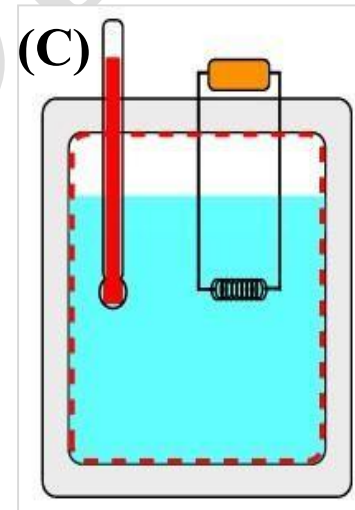
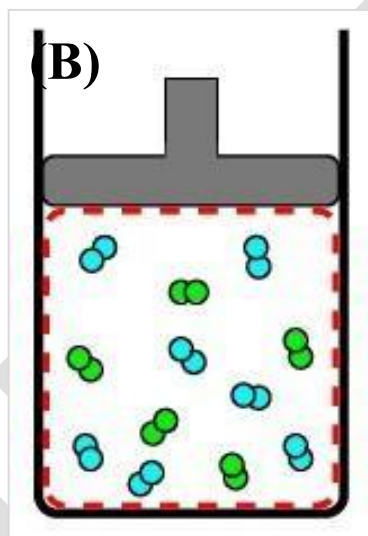
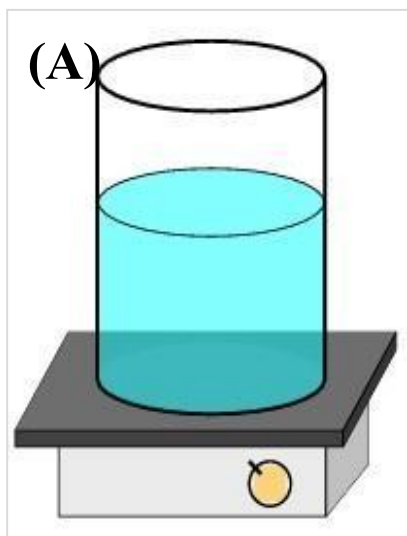
- Any part of the universe which is selected for thermodynamic study is called a **system** and the rest of the Universe in its neighborhood is known as **surroundings** .

- The system is separated from the surroundings by a real or imaginary boundary through which exchange of energy or matter may take place.
- Interactions between the system and its surrounding can be classified as: (i) **Open** , (ii) **Closed** or (iii) **Isolated system**.

1.2.1 Types of System

- A. Open system:** Exchange both energy and matter with its surroundings.
- B. Closed system:** Can exchange energy but not matter with the surroundings.

C. Isolated System: There is no transfer of either energy or matter with the surroundings.



1.3 SAQ-1

Classify the following systems as open, closed or isolated.

1) Nitrogen and hydrogen reacting to form ammonia in a sealed tube.

- 2) Potassium chlorate and manganese dioxide are heated in an unsealed test tube to form potassium chloride and oxygen.
- 3) A glass vial containing conc. Hydrochloric acid is broken inside water in a beaker.
- 4) A glass vial containing hydrochloric acid is broken inside a solution of sodium hydroxide kept in a closed polythene bottle insulated with a cork. **1.5**

Properties of a System

The properties associated with a macroscopic system are called thermodynamic properties or variables

Two main categories:

- 1) **EXTENSIVE PROPERTIES:-** Depend upon the amount/quantity of matter present in the system e.g.mass, volume, energy etc.
- 2) **INTENSIVE PROPERTIES:** depends on characteristics of matter but independent of its amount e.g. pressure, temperature, density, viscosity, surface tension, refractive index, electromotive force, chemical potential, specific heat, etc.

1.4 State functions

Internal energy, a state function, depends only on the present state and NOT on the path by which it arrived at that state.

Example:

The internal energy of 50g of water at 25°C is the same whether the water is cooled from a higher temperature to 25°C or is obtained by melting 50 g of ice and then warming to 25°C

The value of a state function does not depend on the particular history of the sample, only on its present condition. The properties that depend only on the state of the system and not on its past history are called **STATE FUNCTIONS**.

STATE FUNCTION

1. Must have a definite value for a given system in a given state, independent of the history of that system and

2. Change by a definite amount in a given change of state, independent of the path along which that change proceeds.

Thermodynamic process:

The method by which the state of a system is changed is called a 'process'. It can be effected by changing any one of the state variables viz P, T, C etc.

Isothermal process: It is a process carried out at constant temperature $dT=0$

Adiabatic process: It is a process in which no exchange of heat takes place. The temperature of the system may increase or decrease during adiabatic process. $dQ=0$

Isobaric process: Carried out at constant pressure

Isochoric process: Carried out at constant volume

Reversible process:

A process which occurs infinitely slowly, can be reversed along the same path and performs maximum work is referred to as being thermodynamically reversible.

It is a process which takes place infinitesimally slowly so that the system is in thermodynamic equilibrium at any instant of the change.

Since the process is carried out extremely slowly the properties of the system remain virtually unchanged and the direction may be reversed by small change in a variable like temperature, pressure etc.

The driving force is greater than the opposing force only by a infinitesimal quantity and hence the process would require infinite time for completion.

Irreversible process:

It is a process which takes place rapidly or spontaneously so that it is in equilibrium with the surroundings. The driving force differ from the opposing force by a large amount and hence it cannot be reversed unless some external force is applied. All the natural processes are irreversible.

Reversible process:

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	Reversible Process	Irreversible process

1	Driving force and opposing force differ by small amount	Driving force and opposing force differ by a large amount
2	It is a slow process	It is a rapid process
3	The work obtained is more	The work obtained is less
4	It is an imaginary process	It is a real process
5	It consists of many steps	It has only two steps i.e. initial and final
6	It occurs in both directions	It occurs in only one direction
7	It can be reversed by changing thermodynamic variables	It cannot be reversed

SAQ-2:

Which of the following are extensive properties and which are intensive properties?

a. Viscosity

- b. Weight
- c. Mass
- d. Volume of a solid
- e. Temperature
- f. Pressure

1.6 The First law of Thermodynamics:

Define first law of thermodynamics both verbally and by means of a equation

In general, energy can be converted from one form to another. But energy can neither be created nor destroyed: Energy is conserved.

- The total energy lost by a system equals the total energy gained by its surroundings. This is the law of conservation of energy also known as the first law of thermodynamics

Alternatively:

The total energy of a system and its surroundings must remain constant, although it may be changed from one form to another"

Internal Energy

- The total energy of a system is the sum of all the kinetic and potential energies of its component parts. For the system of H_2 and O_2 molecules, the total energy include not only the motions and interactions of the H_2 and O_2 molecules themselves but also of the component nuclei and electrons.
- The total energy is called the **internal energy** of the system
- Because there are so many types of motions and interactions, we cannot determine the exact energy of any system of practical interest. We can, however, measure the change in

internal energy that accompany chemical and physical processes.

- $\Delta E = E_{\text{final}} - E_{\text{initial}}$

In terms of the work performed (w), the heat (q) transferred and the change in internal energy (ΔE) that is stored in the system; first law of Thermodynamics is mathematically expressed as:

$$\Delta E = q - w \quad \text{or} \quad q = \Delta E + w$$

For very small changes: $dE = dq - w$

Where dE is the small change in the internal energy, q is the small amount of heat absorbed and w is the small amount of work done by the system.

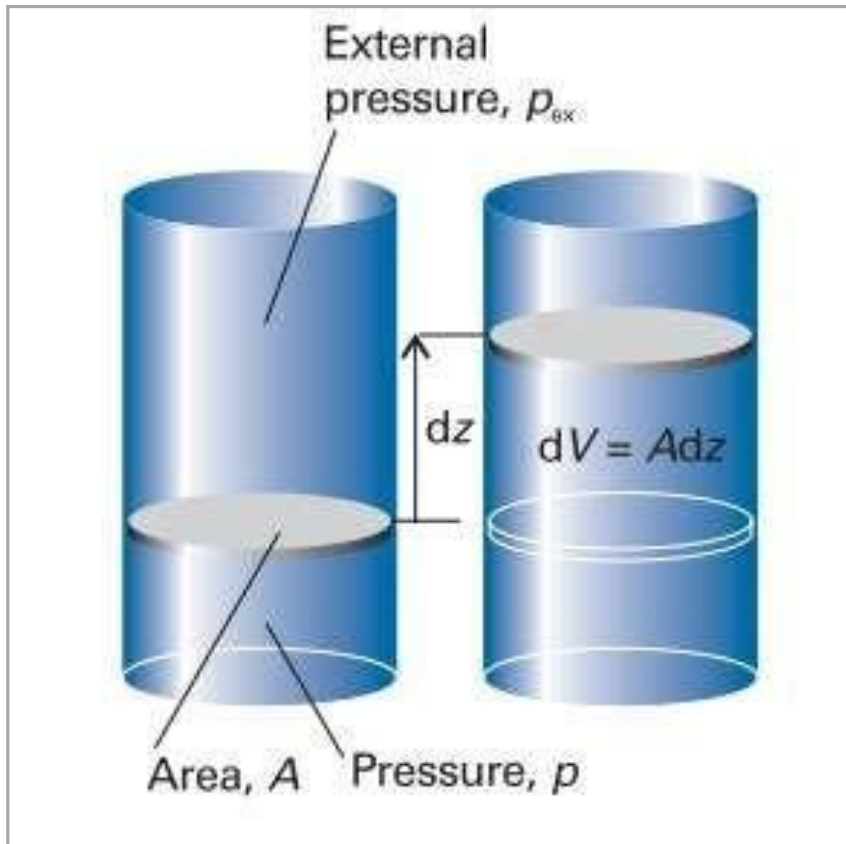
- This implies that "of the total heat, q , absorbed from the surroundings a portion raises the internal energy of the system by ΔE and the rest is used as the work of expansion".
- Heat transfer may occur by conduction, convection or radiation

Summary of sign conventions

	Positive (+)	Negative (-)
Heat (q)	System <u>gains</u> energy	System <u>loses</u> energy
Work (w)	Work done <u>by</u> the system	Work done <u>on</u> the system
Change in internal energy (ΔE)	Internal energy of the system <u>increases</u>	Internal energy of the system <u>decreases</u>

Expansion or compression work :

SIR OTONDE



- The external pressure P_{ex} is equivalent to a weight pressing on the piston, and the force opposing expansion is:

$$\text{work (w)} = F \times d$$

But $F = P_{\text{ex}}A$, then:

$$w = P_{\text{ex}}Adz \Rightarrow w = P_{\text{ex}}\Delta V$$

When a piston of area A moves out through a distance dz it sweeps out a volume $dV = Adz$.

- In expanding, any system does work by pushing back the atmosphere around it; in contracting, any system is the recipient of work done by the atmosphere pressing upon it
- Let the normal external pressure be P_{ex} and the cross-sectional area be A
- Work done by or on the system when the piston moves dz
- $\text{Work} = Mgdz = \frac{Mg}{A} \cdot A \cdot dz = P_{\text{ex}} dV$ ($dV = Adz$)
- For a finite movement of the piston

$$W = \sum P_{\text{ex}} dV = \int_{V_1}^{V_2} P_{\text{ex}} dV$$

$$\text{For constant } P_{\text{ex}} \quad W = P_{\text{ex}} \int_{V_1}^{V_2} dV = P_{\text{ex}} (V_2 - V_1) = P_{\text{ex}} \Delta V$$

ΔV – finite change in volume

What is the magnitude of reversible work, W_{rev} ?

- $W = \int_{V_1}^{V_2} P_{\text{ex}} dV$ and we must express P_{ex} as a function of V
- In a reversible change $P_{\text{ex}} = P \pm dP$ where P is the interior gas pressure
- $W_{\text{rev}} = \int_{V_1}^{V_2} (P \pm dP) dV = \int_{V_1}^{V_2} (PdV \pm dPdV)$
- The term $dPdV$ is an infinitesimal of higher order; no error attends its complete suppression.
- $W_{\text{rev}} = \int_{V_1}^{V_2} PdV$
- For an ideal gas $PV = nRT$ and therefore $P = \frac{nRT}{V}$
- $W_{\text{rev}} = \int_{V_1}^{V_2} \frac{nRT}{V} dV$

In the isothermal change of a fixed quantity of gas

$$W_{\text{rev}} = nRT \int_{V_1}^{V_2} \frac{1}{V} dV = nRT \ln \frac{V_2}{V_1}$$

NOTE: When a system undergoes a given change of state, the maximum amount of recoverable work varies markedly with the path along which the change proceeds.

Question 1

Calculate the work done in kilojoules done during a reaction in which the volume expands from 12.0L to 14.5L against an external pressure of 5.0 atm.

Solution 1

$w = -P\Delta V$ (Expanding system loses energy, which thus have a negative sign)

Given: $P = 5.0 \text{ atm}$, and $\Delta V = (14.5 - 12.0) = 2.5\text{L}$.

$$w = -(5.0 \text{ atm}) \times (2.5\text{L}) = -12.5 \text{ L.atm}$$

$$= -(12.5 \text{ L.atm}) \times \left(101 \frac{\text{J}}{\text{L.atm}}\right)$$

$$= -1.3 \times 10^3 \text{ J}$$

$$= -1.3\text{kJ}$$

$$1 \text{ L.atm} = (1 \text{ L}) \left(\frac{10^{-3} \text{ m}^3}{1 \text{ L}} \right) \left(101 \times 10^3 \frac{\text{kg}}{\text{m.s}^2} \right) = 101 \frac{\text{kg.m}^2}{\text{s}^2} = 101 \text{ J}$$

Question-2:

When one mole of water, at 100 °C and 1 atm. pressure, is converted to steam (at 100 °C) the amount of heat absorbed is 40670 J. Calculate ΔE for the change. **Solution 2:**

Work done in expanding against atmospheric pressure: w
$$= P_{\text{ext}} (V_2 - V_1)$$

Given: V_1 = Volume of one mole of a liquid water
$$= 18 \text{ cm}^3 = 18 \times 10^{-6} \text{ m}^3$$

V_2 = volume of 1 mole of water vapour at 100 °C

From Gas laws and at constant pressure: $V_1/T_1 = V_2/T_2$ Also, at STP (273 K and 1 atm pressure) 1 mole of ideal gas occupies $\approx 0.0224 \text{ m}^3$

Therefore: $V_2 = 0.0224 * (373/273) = 0.0306 \text{ m}^3$

Then; $P_{\text{ext}} = 101325(0.0306 - 18 \times 10^{-6}) = 3099 \text{ J}$ From

First Law: $\Delta E = q - w = (40670 - 3099) = \underline{\underline{37571 \text{ J}}}$.

Question-3:

A balloon is filled with helium is heated and expands, performing 370 J of work on the surrounding gases in the atmosphere. The internal energy of helium increases by 1220 J. How much heat did helium absorb?

Solution 3:

$$\Delta E = q - w$$

$$\text{Given: } w = 370 \text{ J, } \Delta E = 1220 \text{ J}$$

$$1220 \text{ J} = q - 370 \text{ J } \underline{q}$$

$$\underline{\underline{= 1590 \text{ J}}}$$

SAQ-3

Two moles of an ideal gas at 273 K and 101325 Nm^{-2} pressure expand from 0.056 m^3 to 0.28 m^3 at the same temperature.

Determine the work done.

Question-4:

3dm³ of hydrogen initially at STP are expanded isothermally and reversibly to a volume of 4 dm³. Calculate the work done.

Solution-4

Given:

n:- 22.415 dm³ is the volume occupied by 1mole at STP and thus 3dm³ is occupied by = $1 \times 3 / 22.415$ mole at STP

$V_2 = 4 \text{ dm}^3$ and $V_1 = 3 \text{ dm}^3$, $T = 273 \text{ K}$, $R = 8.314 \text{ JK}^{-1}\text{mol}^{-1}$

Using equation 5:
$$W = nRT \ln \left(\frac{V_2}{V_1} \right)$$
$$= \left(\frac{3}{22.415} \right) 8.314 * 273 \ln \left(\frac{4}{3} \right)$$
$$= \underline{\underline{87.39 \text{ J}}}$$

INTERNAL ENERGY OF AN IDEAL GAS

- **Internal energy for an ideal gas is of special interest. Why?**

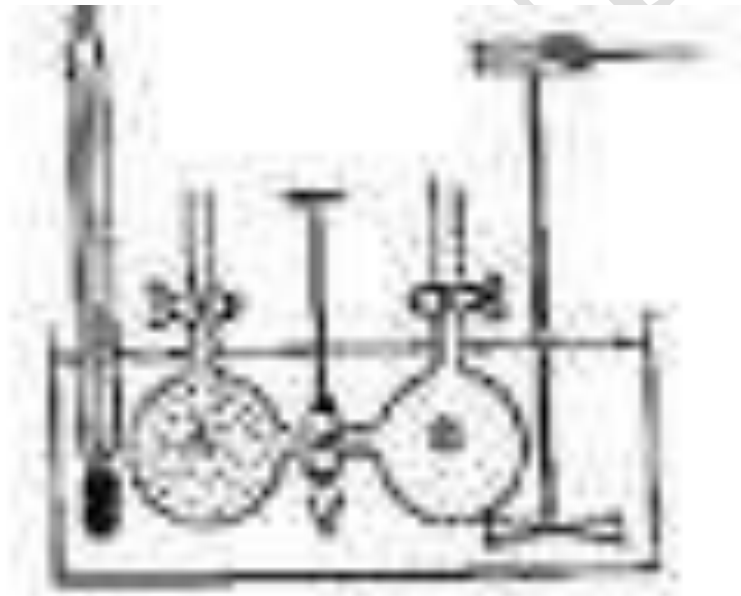


Figure 1 Joule Apparatus

- **Molecules of an ideal gas are not supposed to attract each other. This implies that $q=0$, when the gas is allowed to expand into a vacuum and no work is done since the external pressure is zero i.e. $w=0$.**

INTERNAL ENERGY OF AN IDEAL GAS

- **Thus for isothermal expansion, $\Delta E=0$ and hence at any constant temperature, the internal energy of an ideal gas is independent**

of its volume. Note this could be in a vacuum or not.

INTERNAL ENERGY OF AN IDEAL GAS

This implies that at constant temperature

$\Delta E=0$ and from $q=\Delta E+w$

$$q=w=nRT \ln \frac{v_2}{v_1} = nRT \ln \frac{P_1}{P_2}$$

Question-5:

0.1 mole of an ideal gas is expanded isothermally at 273 K from 3 dm³ to 5 dm³. Determine the energy (q) absorbed from the surroundings.

Solution 5:

Given: $n = 0.1$ mole, $T = 273$, $V_1 = 3$ dm³ and $V_2 = 5$ dm³

Using equation (7) $q = 0.1 * 8.314 *$

$$273 \ln (5/3) =$$

$$\underline{\underline{115.94 \text{ J}}}$$

Derivation of T – P – V relations for adiabatic and reversible process

- An adiabatic process occurs if there is no exchange of heat (q) between the system and surroundings. That is, $q = 0$ in an adiabatic process.
- In an adiabatic process if work is done by the system its temperature decreases, if work is done on the system its temperature increases.

Derivation of T – P – V relations for adiabatic and reversible process

- It is important to review the concept of heat capacities at constant pressure and at constant volume since the heat capacities will feature in the calculations..

Derivation of T – P – V relations for adiabatic

- Molar heat capacity, C , is defined as the heat required to raise the temperature of one mole of the system by one degree.

and reversible process

Derivation of T – P – V relations for adiabatic

In differential form this is expressed as $C = \frac{\delta q}{dT}$. The molar heat capacity at constant volume is

therefore $C_v = \frac{\delta q_v}{dT}$ while for constant pressure is $C_p = \frac{\delta q_p}{dT}$. We note that $\delta q_v = \Delta E$ in

which case then we have $C_v = \frac{dE}{dT}$ or $dE = C_v dT$ Integration yields $\int_{E_1}^{E_2} dE = \int_{T_1}^{T_2} C_v dT$

leading to $\Delta E = E_2 - E_1 = \int_{T_1}^{T_2} C_v dT = C_v(T_2 - T_1)$ if C_v is independent of temperature.

Similarly $\delta q_p = \Delta H$ implying $C_p = \frac{dH}{dT}$ Meaning $dH = C_p dT$ Integration yields

$\Delta H = H_2 - H_1 = \int_{T_1}^{T_2} C_p dT = C_p(T_2 - T_1)$ when C_p is independent of temperature.

Derivation of T – P – V relations for adiabatic and reversible process

Now one wants to find a relationship between C_v and C_p . From the First Law of thermodynamics $\delta q = \delta w + dE$. And at constant pressure, one has $\delta q_p = dE + PdV$

Noting that for one mole $PV = RT$ then $PdV = RdT$ and this implies that $\delta q_p = dE + RdT$

Dividing throughout by dT yields $\frac{\delta q_p}{dT} = \frac{dE}{dT} + R$ and thus $C_p = C_v + R$

Derivation of T – P – V relations for adiabatic and reversible process

Purpose of derivation: The relation $w = \Delta E = C_V (T_2 - T_1)$ is used to calculate the desired quantities when T_2 and T_1 are known. When instead of T_2 the values of P_1 , V_1 , T_1 and P_2 , V_2 are given, we need relations between T , P and V .

1. T-V RELATION FOR AN ADIABATIC PROCESS

$$C_V dT = -PdV \text{ and for gases } P = \frac{nRT}{V}$$

$$C_V dT = -\frac{nRT}{V} dV \text{ or } \frac{dT}{T} = -\frac{nR}{C_V} \frac{dV}{V}$$

1. T-V RELATION FOR AN ADIABATIC PROCESS

Or $d \ln T = -d \ln(V)^{\gamma-1}$ where $\gamma = \frac{C_p}{C_v}$ and $\gamma - 1 = \frac{nR}{C_v} = (C_p - C_v)/C_v$

On integrating between the limits $T=T_1$, $T=T_2$ and $V=V_1$, $V=V_2$ we get $\ln \frac{T_2}{T_1} = \ln \left(\frac{V_1}{V_2} \right)^{\gamma-1}$

On taking antilog, we shall get $\frac{T_2}{T_1} = \left(\frac{V_1}{V_2} \right)^{\gamma-1}$ or $T_2 = T_1 \left(\frac{V_1}{V_2} \right)^{\gamma-1}$

2. $T - P$ RELATION FOR OR AN ADIABATIC PROCESS

$$C_v dT = -PdV \quad (i)$$

For an ideal gas $PV = nRT$ and $PdV + VdP = nRdT$

$$\text{or } -PdV = VdP - nRdT = \frac{nRT}{P} dP - nRdT \quad (ii)$$

From equations (i) and (ii) , you get

$$(C_v + nR) \frac{dT}{T} = nR \frac{dP}{P}$$

2. $T - P$ RELATION FOR OR AN ADIABATIC PROCESS

$$\text{or } d \ln T = \frac{nR}{C_v + nR} d \ln P = d \ln (P)^{(\gamma - 1)/\gamma}$$

$$\text{Where } \gamma = \frac{C_p}{C_v} \text{ and } \gamma - 1 = \frac{nR}{C_v} = \frac{(C_p - C_v)}{C_v}$$

On integrating between the limits $T=T_1$, $T=T_2$, and $P=P_1$, $P=P_2$, you get

$$\ln \frac{T_2}{T_1} = \ln \left(\frac{P_2}{P_1} \right)^{(\gamma - 1)/\gamma}$$

2. $T - P$ RELATION FOR OR AN ADIABATIC PROCESS

On taking antilog, you shall get

$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{(\gamma-1)/\gamma}$$

$$T_2 = T_1 \left(\frac{P_2}{P_1}\right)^{(\gamma-1)/\gamma}$$

3.

P – V RELATION FOR AN ADIABATIC PROCESS

$$C_v dT = -P dV$$

For an ideal gas $PV = nRT$ and $PdV + VdP = nRdT$

$$\frac{nRT}{V} dV + \frac{nRT}{P} dP = nRdT$$

$$\text{Or } dT = T \frac{dV}{V} + T \frac{dP}{P} = T(d\ln V + d\ln P)$$

On substituting dT into equation (i) you get

$$C_v \times T(d\ln V + d\ln P) = -P dV = -\frac{nRT}{V} dV = -nRT d\ln V$$

3. *P – V RELATION FOR AN ADIABATIC PROCESS*

Or $(C_v + nRT)d\ln V = -C_v d\ln P$ or $d\ln(V)^\gamma = -d\ln P$

On integrating between the limits $V=V_1$, $V=V_2$, and $P=P_1$, $P=P_2$, then taking antilog you shall get $(\frac{V_2}{V_1})^\gamma = \frac{P_1}{P_2}$ and $P_1 V_1^\gamma = P_2 V_2^\gamma = \text{constant}$

$PV^\gamma = \text{constant}$

Question-6:

Two moles of an ideal monatomic gas at 300 K are compressed adiabatically to one quarter of the original volume. What is the temperature of the gas after compression.

SOLUTION:

For a monatomic gas;

$$C_v = \frac{3}{2}R$$

Then we are given;

$$T_1 = 300 \text{ K}, V_2 = V_1/4, T_2 = ?$$

Equation (24)

$C_V \ln T_2/T_1 = -R \ln V_1/V_2$ Substitution gives

$$\frac{3}{2}R \ln T_2/300 = -R \ln V_1/4V_1 \ln T_2/300 \\ = \frac{2}{3} \ln 4$$

Hence: $T_2 = 755.95 \text{ K}$ Ans

Question-7:

To what pressure must a given volume of Helium originally at 100°C and 1 atm. Pressure be adiabatically compressed in order to raise its temperature to 400°C ?

Solution:

$$P_1 = 1 \text{ atm} = 101\,325 \text{ NM}^{-2}$$

$$T_2 = 400 \text{ }^\circ\text{C} = 673 \text{ K}$$

$$T_1 = 100 \text{ }^\circ\text{C} = 373 \text{ K}$$

$$P_2 = ?$$

$C_p = C_v + R$ and $C_v = 3/2 R$ for a monatomic gas.

$$\therefore C_p = 5/2 R$$

Then

$$P_1(T_2)^{C_p/R} = P_2(T_1)^{C_p/R}$$

Substitution gives

$$101325 \times (673)^{5/2} = P_2(373)^{5/2}$$

$$P_2 = 443080.05 \text{ NM}^{-2}$$

$$= \underline{\underline{4.37 \text{ atm}}}$$

Tutorial Question. 1

a) Explain clearly what is meant by a thermodynamically reversible process. Solution:

a) A thermodynamic reversible process occurs in infinitely slowly, can be reversed along the same path and performs maximum work.

b) Explain the thermodynamic meaning of a system, distinguishing among the open, closed and isolated systems.

Solution:

b) A system is part of the universe, the properties of which are under investigations and has a definite boundary.

□ **Open system:** Exchange both energy and matter with surroundings.

□ **Closed system:** Exchange energy with surroundings, matter NOT exchanged.

- **Isolated system**: No transfer of either energy or matter with surroundings

Tutorial Question. 1...

- c) One mole of an ideal gas at 25 °C and a pressure of $10 \times 10^5 \text{ Nm}^{-2}$ expands isothermally to a pressure of $1 \times 10^5 \text{ Nm}^{-2}$
- Calculate the final volume, w and q if the expansion were carried out reversibly.
 - Calculate w and q for an irreversible expansion

Solution:

c) (i). $PV = nRT \Rightarrow V = \frac{nRT}{P} = \frac{1 \text{ mol} \times 8.314 \text{ JK}^{-1}\text{mol}^{-1} \times 298 \text{ K}}{1 \times 10^5 \text{ Nm}^{-2}} = 0.024 \text{ m}^3$

$$w = P\Delta V = nRT \ln \frac{P_1}{P_2} = 1 \times 298 \times 8.314 \ln \frac{10 \times 10^5}{1 \times 10^5} = 5704.8 \text{ J}$$

First law of thermodynamics: $\Delta E = q - w$, $\Delta E = 0: \Rightarrow q = w$
 $q = 5704.8 \text{ J}$

Solution:

c) (ii). $w = 1 \times 10^5 \times 8.314 \times 298 \left[\frac{1}{1 \times 10^5} - \frac{1}{10 \times 10^5} \right] = 2229.8 \text{ J}$

First law of thermodynamics: $\Delta E = q - w$, $\Delta E = 0: \Rightarrow q = w$
 $q = 2229.8 \text{ J}$

Tutorial Question. 2

A sample of helium at 1.0 atm pressure and 25 °C expands reversibly and adiabatically from 1.00 L to 4.00 L. Determine the final temperature and work done during expansion. (Given $C_V = \frac{3}{2}R$)

Solution:

Given: $P_1 = 1 \text{ atm}$, $T_1 = 298 \text{ K}$ and $V_1 = 1.0 \text{ L}$

$$\therefore n = \frac{P_1 V_1}{RT_1} = \frac{1 \times 1}{0.0821 \times 298} = 0.041 \text{ moles}$$

$$\text{Then: } C_V \ln \frac{T_2}{T_1} = -R \ln \left(\frac{V_2}{V_1} \right)$$

$$\frac{3}{2} R \ln \left(\frac{T_2}{298} \right) = -R \ln \left(\frac{4}{1} \right)$$

$$T_2 = 118.26 \text{ K}$$

$$w = -n C_V (T_2 - T_1)$$

$$= -0.041 \times \frac{3}{2} R (118.26 - 298)$$

$$= 91.903 \text{ J.}$$

SAQ-3

- 1) To what pressure must a given volume of Helium originally at 100°C and 1 atm. Pressure be adiabatically compressed in order to raise its temperature to 400°C ?
- 2) At 0°C and 1 atm. Pressure the volume of 1 mole of an ideal monatomic gas is 22.415 litres. The gas is expanded until its pressure is 0.4 atm. by a reversible adiabatic process. Determine the final volume.
- 3) The volume of a sample of an ideal monatomic gas at 0°C is 44.83 litres. To what volume must the gas be compressed adiabatically so as to attain a temperature of 30°C

Summary

- Energy cannot be created or destroyed but can be transformed from one form to another.
- The heat absorbed by the system is partly stored within it as internal energy and partly spent on doing work on the surroundings.
- Heat absorbed by the system is taken to be positive.
- Internal energy is a state function.
- Transformations of systems can take place reversibly or irreversibly.
- Transformations taking place in an infinite number of steps permitting equilibrium, to prevail at every stage are called reversible transformations. When the transformation is rapid and sudden, it is said to be spontaneous or irreversible.

Summary (cont.)

- Work is calculated as a product of the pressure against which work is done and the change in volume. For an irreversible process the pressure (external) is often constant. For a reversible process, the pressure changes during the course of the process.
- The heat absorbed by a system at constant volume is equal to the internal energy change while that absorbed by the system at constant pressure is the enthalpy change.
- The temperature coefficient of the internal energy change at constant volume is the heat capacity at constant volume and the temperature coefficient of the enthalpy change at constant pressure is the heat capacity at constant pressure.
- An adiabatic change is that which can be carried out reversibly such that no heat enters or leaves the system

Enthalpy at Constant Volume/ Pressure

Using the symbol q to represent transferred heat and work $w = P\Delta V$; the total internal energy change of the system can be represented as:

$$\Delta E = q - w = q - P\Delta V$$

Rewrite: $q = \Delta E + P\Delta V$

Constant volume:

Reactions carried out in a closed container are at a constant volume.

Therefore; $\Delta V = 0$, then $q_v = \Delta E$ (at constant volume).

Constant pressure:

Reactions carried out in open flask keeping the pressure constant and allowing the volume of the system to change; energy change in

the system is due to both heat transfer and PV work. $q_p = \Delta E + P\Delta V$ (at constant pressure).

Enthalpy at Constant Volume/ Pressure con't

- Most physical and chemical changes, including those in living systems, take place under the essentially constant pressure of earth's atmosphere.
- Chemists use the symbol **H**, and a special name, **enthalpy** (enthalpein to 'warm') in dealing with heat absorbed or released under constant pressure. Therefore q_p can be substituted by ΔH ;

$$\Delta H = \Delta E + P\Delta V$$

*enthalpy a state function whose value only depends on the current state of the system, not on the path taken to arrive.

i. e. $\Delta H = H_{\text{final}} - H_{\text{initial}}$

In physical terms, enthalpy is viewed as the heat content of a system.

□ Enthalpy is an extensive property

The magnitude of ΔH is directly proportional to the amount of reactants consumed in the process

□ The enthalpy change for a reaction is equal in magnitude but opposite in sign to ΔH for the reverse reaction

□ The enthalpy change for a reaction depends on the state of the reactants and products

□ The physical states of the reactants and products need to be considered when calculating ΔH

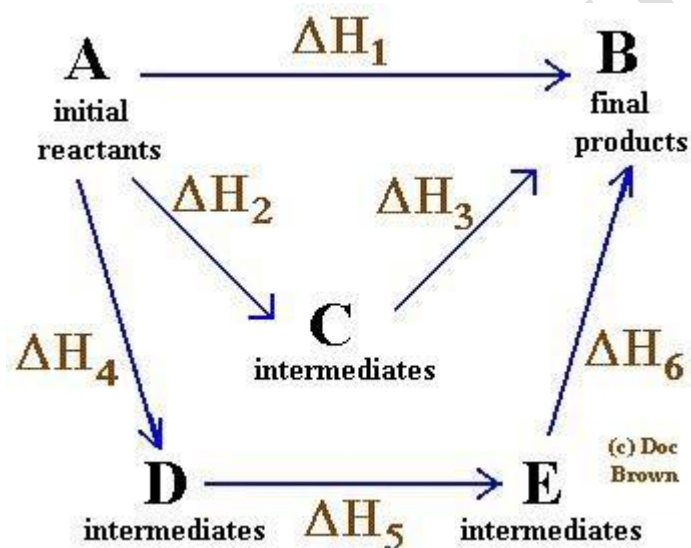
HESS'S LAW

- Many enthalpies of reactions have been measured and tabulated. It is often possible to calculate ΔH values of other reactions.
- Because enthalpy is a state function, the enthalpy change, ΔH , associated with any chemical process depends only on the amount of matter that undergoes change and on the nature of the initial state of reactants and the final state of reactants

HESS'S LAW

STATES THAT “If a reaction is carried out in a series of steps, ΔH for the reaction will be equal to the sum of the enthalpy changes of the individual steps”

The overall enthalpy change for the process is independent of the number of steps or the particular nature of the path by which the reaction is carried out.

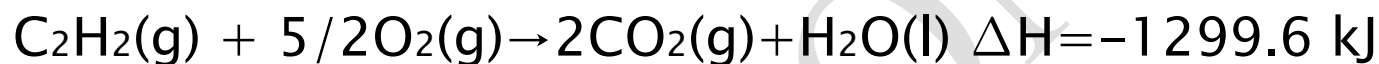


Example

Calculate ΔH for the reaction



Given the following reactions and their respective enthalpy changes:



Solution

- 1) Because the target equation has C_2H_2 as a product, we turn the first equation around; the sign of ΔH is therefore changed

- 2) Because the target equation has 2C(s) as a reactant, we multiply the second equation and its ΔH by 2
- 3) Because the target equation has H_2 as a reactant, we keep the third equation as it is. We then add the three equations and their enthalpy in accordance with Hess's law.
- 4) When the equations are added, there are 2CO_2 , $5/2\text{O}_2$ and H_2O on both sides of the arrow. These are cancelled in writing the net equation.

Enthalpies of Formation

- We can calculate the enthalpy changes from a great many reactions from a few tabulated ΔH values.

Extensive tables exist for

- **Enthalpies of vaporization** - ΔH_{vap} for converting liquid to gas
- **Enthalpies of fusion** - ΔH_{f} for melting solids
- **Enthalpies of combustion** - ΔH_{c} for combusting a substance in oxygen and so on

A particularly important process for tabulating thermochemical data is the **formation of a compound from its constituent elements**.

- † The enthalpy change associated with this process is called the **ENTHALPY OF FORMATION (or HEAT OF FORMATION)** labeled ΔH_{f}

† The **MAGNITUDE OF ANY ENTHALPY CHANGE** depends on the conditions of **TEMPERATURE, PRESSURE** and **STATE**

(gas, liquid or solid, crystalline form) of the reactants and products.

† The **STANDARD STATE** for a substance is the form most stable at a particular temperature of interest and at standard atmospheric pressure.

- When a reaction occurs with all reactants and products in their standard states, we refer to its enthalpy change as the

STANDARD ENTHALPY OF REACTION, ΔH° . The temperature usually chosen for tabulating data is **298 K (25°C)**

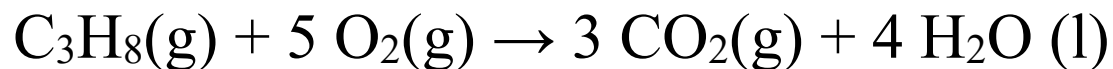
- The **STANDARD ENTHALPY OF FORMATION** of a compound, ΔH°_f , is the **change in enthalpy that**

accompanies the formation of 1 mol of that substance from its elements in their standard states.

- $2\text{C}(\text{graphite}) + 3\text{H}_2(\text{g}) + \frac{1}{2}\text{O}_2(\text{g}) \rightarrow \text{C}_2\text{H}_5\text{OH}(\text{l})$ $\Delta H^\circ_f = -277.7 \text{ kJ}$
(O_2 most stable NOT O or O_3 at 25°C , Elemental source of carbon is graphite NOT diamond)
- We can determine the standard enthalpy change for any reaction, $\Delta H^\circ_{\text{rxn}}$, by using standard enthalpies of formation and Hess's law

$$\Delta H^\circ_{\text{rxn}} = \sum n \Delta H^\circ_f(\text{products}) - \sum m \Delta H^\circ_f(\text{reactants})$$

- For example, we can calculate ΔH° for the combustion of propane from the ΔH°_f (reactants values of the products and reactants:



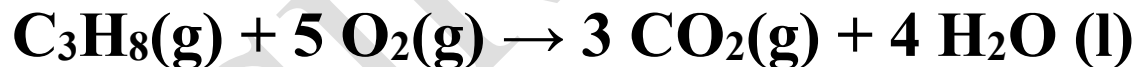
$$\Delta H^\circ_{\text{rxn}} = [3\Delta H^\circ_{\text{f}}(\text{CO}_2) + 4 \Delta H^\circ_{\text{f}}(\text{H}_2\text{O})] - [\Delta H^\circ_{\text{f}}(\text{C}_3\text{H}_8) + 5\Delta H^\circ_{\text{f}}(\text{O}_2)]$$

Because $\text{O}_2(\text{g})$ is the most stable form of the element, $\Delta H^\circ_{\text{f}}(\text{O}_2) = 0$. The other enthalpies of formation are obtained from data tables.

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Standard Enthalpy of Formation ($\Delta H^\circ_{\text{f}}$)

SUBSTANCE	$\Delta H^\circ_{\text{f}0298}, \text{kJ}$
$\text{H}_2\text{O}(\text{l})$	- 285.56
$\text{CO}_2(\text{g})$	- 393.14
$\text{H}_2\text{S}(\text{g})$	- 17.49
$\text{CH}_4(\text{g})$	- 74.78
$\text{CH}_3\text{OH}(\text{l})$	- 238.34
$\text{C}_3\text{H}_8(\text{g})$	

Tutorial Question

Calculate the heat of reaction for $\text{CO(g)} + \text{H}_2\text{O(l)} \rightarrow \text{CO}_{2\text{(g)}} + \text{H}_{2\text{(g)}}$ at 293 K from the given table of heats of formation.

SUBSTANCE	$\Delta H_{f0293, \text{kJ}}$
$\text{H}_2\text{O(l)}$	- 286.86
CO(g)	- 110.88
$\text{CO}_{2\text{(g)}}$	- 394.8

Solution

$$\begin{aligned}\Delta H_{r298}^0 &= [\Delta H_f^0(\text{CO}_2) + (\Delta H_f^0(\text{H}_2))] - [\Delta H_f^0(\text{CO}) + \Delta H_f^0(\text{H}_2\text{O})] \\ &= [(-394.8) + (0)] - [(-110.88) + (-286.86)] \\ &= -394.8 + 397.74 \\ &= \underline{\underline{2.94 \text{ kJ}}}\end{aligned}$$

CALORIMETRY

- The value of ΔH is determined experimentally by measuring the heat flow accompanying a reaction at constant pressure. When heat flows into or out of a substance, the temperature of the substance changes.

- Experimentally we can determine the heat flow associated with a chemical reaction by measuring the temperature change it produces. The measurement of heat flow is called **CALORIMETRY**; an apparatus that measures heat flow is called **CALORIMETER**.

- The calorimeter consists of a container in which the reaction releasing (or absorbing) heat is caused to occur. The evolved heat flows into a

known amount of water or other material which surrounds the reaction vessel.

- Knowing the heat capacity of this absorbing jacket, and measuring the rise in its temperature, we may use;

$$Q = m_A c_A \Delta T$$

where Q is the heat, m_A is the mass, c_A is the heat capacity and ΔT is the change in temperature.

The calorimeter used may be of **constant volume** or **constant pressure** type.

CALORIMETRY Calibration

- The temperature change experienced by a body when it absorbs a certain amount of heat is determined by its heat capacity. ***Heat capacity of an object is the amount of heat required to raise its temperature by 1 K or (1°C).*** For pure substances, the heat capacity of 1 mol of a substance is referred to as molar heat capacity. And for 1 g of a substance, specific heat capacity or specific heat.

$$\text{Specific heat} = \frac{\text{quantity of heat transferred}}{(\text{grams of substance}) \times (\text{temperature change})}$$

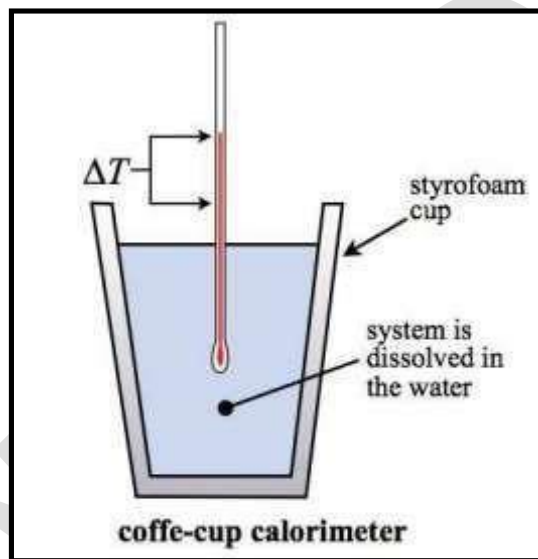
$$= \frac{q}{m \times \Delta T}$$

Constant Pressure Calorimetry

Recall: $\Delta H = q_p$

Reaction occurs under constant pressure of the atmosphere Because the calorimeter prevents the gain or loss of heat from its surroundings,

the heat released by the reaction $-q_{\text{rxn}} = \text{heat gained by solution}$
 $q_{\text{soln}} \quad q_{\text{rxn}} = -q_{\text{soln}} = -(\text{specific heat of solution}) \times (\text{gram of solution}) \times \Delta T$



= 2.94 kJ

Example

When a student mixes 50 mL of 1.0 M HCl and 50 mL of 1.0 M NaOH in a polystyrene cup calorimeter, the temperature of the resultant solution increases from

21.0°C to 27.5°C. Calculate the enthalpy change for the reaction, assuming that the calorimeter loses only a negligible quantity of heat, that the total volume of the solution is 100 mL, that its density is 1.0 g/mL; and that its specific heat is 4.18 J/g-K.

Answer

Mass of solution = (100 mL)(1.0g/mL) = 100 g

Temperature change (ΔT) = 27.5 °C – 21.0 °C = 6.5 K (reaction

exothermic) $q_{\text{rxn}} = -(100 \text{ g})(4.18 \text{ J/gK})(6.5 \text{ K}) = -2700 \text{ J} = -2.7 \text{ kJ}$

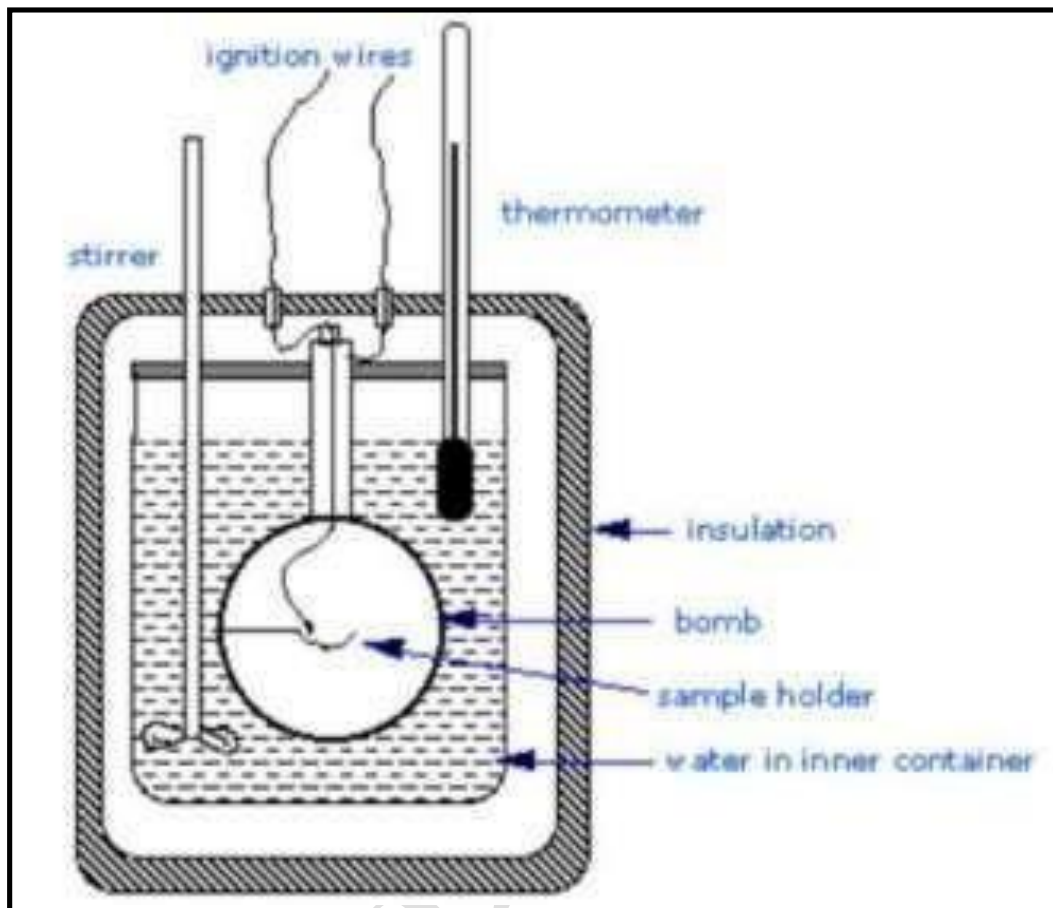
Process occurs at constant pressure: $\Delta H = q_p = -2.7 \text{ kJ}$ moles of HCl or NaOH = [(50mL) x 1.0 mol]/1000mL = 0.050 mol

Enthalpy change per mole of HCl or NaOH

$\Delta H = -2.7 \text{ kJ} / 0.050 \text{ mol} = -54 \text{ kJ/mol}$. **Tutorial question**

A 3.5 g sample of quinine $\text{C}_6\text{H}_4\text{O}_2$ was burned in a calorimeter with a heat capacity of $3.43 \text{ kJ/}^\circ\text{C}$ and contained 1.2 kg of water. The temperature of the calorimeter and water increased from 19.0°C to 28.0°C . What is the molar enthalpy of combustion of quinine? (mm quinine = 108.09 g/mol ; $C_{\text{soln}} = 4.18 \text{ J/g}^\circ\text{C}$)

Bomb Calorimetry (Constant – Volume Calorimetry):



A compound, usually an organic compound, is allowed to react completely with excess oxygen.

After the sample has been placed in the bomb, the bomb is sealed and pressurized with oxygen and placed in an accurately measured quantity of water. The heat evolved in the combustion of the sample is absorbed by its surroundings, namely, the calorimeter contents

- To know the heat capacity of the calorimeter, C_{cal} , we burn a sample that gives off a known quantity of heat. Combustion of 1 g

of benzoic acid, $C_7H_6O_2$, produces 26.38 kJ of heat and if the temperature increase is 5.022°C , then:

$$C_{\text{cal}} = 26.38 \text{ kJ}/5.022^\circ\text{C} = \underline{\underline{5.253 \text{ kJ}/^\circ\text{C}}}$$

For other reactions:

$$q^{\text{evolved}} = -C_{\text{calorimeter}} \times \Delta T$$

Hydrazine, N_2H_4 and its derivatives are widely used as rocket fuels. The combustion of hydrazine with oxygen produces $N_2(g)$ and $H_2O(g)$:



When 1.00 g of hydrazine is burned in a bomb calorimeter, the temperature increases by 3.51°C (heat capacity 5.510 kJ/°C)

- i) What is the quantity of heat evolved
- ii) What amount of heat is evolved upon combustion of a mole of N_2H_4 ?

It may be noted in this case, the volume remains constant. Hence, the enthalpy of reaction gives a measure of q_v $q_p = q_v$

+ ΔnRT

Tutorial Question:

0.50 g of benzoic acid was subjected to combustion in a bomb calorimeter when the temperature of the calorimeter system (including water) was found to rise by 0.55°C . Calculate the enthalpy of combustion of benzoic acid (i) at constant volume (ii) At constant pressure.

The thermal capacity of the calorimeter system including water was found to be 23.85 kJ K^{-1} .

The Effect of Temperature on Enthalpy of Reaction
(The Kirchhoff's Equation):

- Reactions are carried out at various temperatures and it is necessary to determine heats of reaction at temperatures other than those calculated from heats of formation or obtained calorimetrically.
- Consider any process represented thus:



- The enthalpy of reaction at constant pressure

$$\Delta H = H_B - H_A$$

- In order to determine the variation of ΔH with temperature this expression is differentiated with respect to temperature at constant pressure. The result is;

- $$\left(\frac{\partial \Delta H}{\partial T}\right)_p = \left(\frac{\partial H_B}{\partial T}\right)_p - \left(\frac{\partial H_A}{\partial T}\right)_p$$

- By definition
$$C_p = \left(\frac{\partial H}{\partial T}\right)_p$$

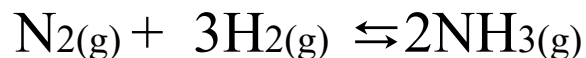
- Therefore
$$\left(\frac{\partial \Delta H}{\partial T}\right)_p = C_{p(B)} - C_{p(A)} = \Delta C_p$$

- Hence its necessary to know the difference in heat capacities of the products and reactants in order to determine the variation of heat of reaction with temperature. This is known as KIRCHOFF'S EQUATION.
- For a small range of temperature ΔC_p can be taken as constant and
- $d(\Delta H) = \Delta C_p dT$
- $\int_{\Delta H_1}^{\Delta H_2} \Delta H = \int_{T_1}^{T_2} \Delta C_p dT$

- If ΔC_p is constant, then
- $\Delta H_2 - \Delta H_1 = \Delta C_p(T_2 - T_1)$
- Re-arranging:
- $\Delta H_2 = \Delta H_1 + \Delta C_p(T_2 - T_1)$
- Where ΔH_2 and ΔH_1 are the heat of reaction at T_2 and T_1 respectively

The Kirchhoff's Equation (cont....):

Question: Consider the reaction below,



the value of ΔH_{298}^0 is - 92.29 kJ at 25 °C. The molar heat capacities at constant pressure of nitrogen, hydrogen and ammonia are given as:

$$C_{H_2} = 29.038 + 0.836 \times 10^{-3}T + 20.097 \times 10^{-7}T^2 \text{ JK}^{-1} \text{ mol}^{-1}$$

$$C_{N_2} = 26.957 + 5.906 \times 10^{-3}T + 3.373 \times 10^{-7}T^2 \text{ JK}^{-1} \text{ mol}^{-1}$$

$$C_{NH_3} = 25.870 + 32.968 \times 10^{-3}T + 30.430 \times 10^{-7}T^2 \text{ JK}^{-1} \text{ mol}^{-1} \quad \text{Calculate}$$

the standard enthalpy of reaction at 125 °C

Solution: The value of ΔC_p is : $\Delta C_p = 2C_{NH_3} - C_{N_2} - 3C_{H_2}$

$$= -62.332 + 62.537 \times 10^{-3}T - 117.79 \times 10^{-7}T^2 \text{ JK}^{-1} \text{ mol}^{-1}$$

Integral part: $\int_{T_1}^{T_2} C_p dT = 62.332(T_2 - T_1) + \frac{62.537 \times 10^{-3}}{2}(T_2^2 - T_1^2) - \frac{117.79 \times 10^{-7}}{3}(T_2^3 - T_1^3)$

$$\int_{T_1}^{T_2} C_p dT = 62.332(T_2 - T_1) +$$

$$39.25 \times 10^{-5} \ln \frac{T_2}{T_1} = 62.332(298K - 298K) +$$

$$= -92.294 - 4.02 \text{ kJ} = \underline{\underline{-96.314 \text{ kJ}}}$$

$$+ \frac{1}{2} \int_{T_1}^{T_2} \frac{C_p}{T^2} dT$$

$$=$$

$$\text{Giving: } 31.271 \times 10^{-5}$$

$$+ \frac{1}{2} \int_{T_1}^{T_2} \frac{C_p}{T^2} dT$$

$$4.02 \text{ kJ}$$

The Kirchhoff's Equation (cont.):

Example:

The standard heat of formation of liquid water is -286.86 kJ at 25 °C. Calculate the value at 100 °C. The molar heat capacities at constant pressure in the given temperature range of H_{2(g)}, O_{2(g)} and H₂O_(l) are 28.84, 29.27 and 75.6 JK⁻¹mol⁻¹, respectively.

Solution:

Heat of formation of water can be written as:

$$\begin{aligned} \text{H}_{2(\text{g})} + \frac{1}{2}\text{O}_{2(\text{g})} &\rightarrow \text{H}_2\text{O}_{(\text{l})} & \Delta H^\circ_{25^\circ\text{C}} &= -286.86 \text{ kJ} \\ \Delta C_p &= C_{p \text{ Products}} - C_{p \text{ Reactants}} \\ &= 75.6 - [28.84 + (\frac{1}{2} \times 29.27)] = 32.125 \text{ J K}^{-1} \end{aligned}$$

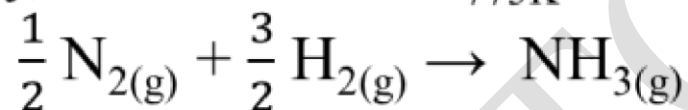
From Kirchoff's law;

$$\begin{aligned} \Delta H_2 &= \Delta H_1 + \Delta C_p (T_2 - T_1) \\ &= -286.86 + \frac{32.125}{1000} (100 - 25) = \underline{\underline{-284.451 \text{ kJ.}}} \end{aligned}$$

The Kirchhoff's Equation (cont.):

SAQ:

The standard heat of formation of gaseous ammonia is -46.28 kJ/mol at 298 K . Given that at 298 K , the constant-pressure heat capacities for $\text{H}_{2(\text{g})}$, $\text{N}_{2(\text{g})}$ and $\text{NH}_{3(\text{g})}$ are 28.84 , 29.23 and $35.2 \text{ JK}^{-1}\text{mol}^{-1}$, respectively, determine $\Delta H^\circ_{773\text{K}}$ for the reaction:



Solution:

For the reaction at 298 K:

$$\begin{aligned}\Delta C_p &= C_{p \text{ Products}} - C_{p \text{ Reactants}} \\ &= 35.2 - \left[\left(\frac{1}{2} \times 29.23 \right) + \left(\frac{3}{2} \times 28.84 \right) \right] = -22.68 \text{ J K}^{-1}\end{aligned}$$

From Kirchoff's law;

$$\begin{aligned}\Delta H_2 &= \Delta H_1 + \Delta C_p (T_2 - T_1) \\ &= -46.28 + \frac{-22.68}{1000} (773 - 298) = \underline{\underline{-13.58 \text{ kJ}}}.\end{aligned}$$

Tutorial Question – Kirchoff's Equation:

Given the following information



Substance	$C_p(\text{JK}^{-1}\text{mol}^{-1})$
$\text{CO}_{(\text{g})}$	$26.6 + 0.76 \times 10^{-2}T$
$\text{CO}_{2(\text{g})}$	$26.4 + 4.26 \times 10^{-2}T$
$\text{O}_{2(\text{g})}$	$25.6 + 1.32 \times 10^{-2}T$

Determine ΔH for the formation of $\text{CO}_{2(\text{g})}$ at 398K.

Answer: $\Delta H = \underline{-283.62 \text{ kJ}}$

Summary:

- The relationship between enthalpy of reaction at constant pressure and at constant volume is governed by the number of moles of the reactants and the number of moles of products.
- The definitions of various forms of enthalpy are dependent on the type of change involved.
- Hess's law of constant heat summation provides a tool for calculating the enthalpy changes by consideration of alternative sequences of reactions.
- The effect of temperature on the enthalpy change of reactions is related to the molar heat capacities of the reactants and products in a relationship defined by the Kirchhoff's equation.

THE SECOND LAW OF THERMODYNAMICS

Introduction

- For a particular process or change, the first law helps us balance the books, so to speak, on the **heat released, work done** and so forth. [$q = \Delta E + w$]
- However, it says nothing about whether the process or change we specify can in fact occur.

Introduction

- That question is encompassed in the **second law of thermodynamics**.

Con't:

Examples of Spontaneous Processes

- If you drop a brick, it falls to the floor
- A shiny nail left outdoors eventually rusts
- Expansion of an ideal gas into an evacuated space.
Reverse not spontaneous

Introduction

- If we add solid KCl to a glass of water at room temperature, we feel the solution growing colder as the salt dissolves. Endothermic and yet spontaneous.
- At a temperature of $0\text{ }^{\circ}\text{C}$ ice spontaneously absorbs heat from the surrounding to turn into a liquid.

Con't:

Which of the following processes are spontaneous? Which ones are non-spontaneous?

- a) Diffusion of perfume molecules from one side of a room to the other side.
- b) Heat flow from a cold object to a hot object.

Introduction

- c) Decomposition of rust ($\text{Fe}_2\text{O}_3 \cdot \text{H}_2\text{O}$) to iron, oxygen and water
- d) Decomposition of solid CaCO_3 to solid CaO and gaseous CO_2 at 25°C and 1 atm pressure.

(a) Spontaneous while (b), (c) and (d) are nonspontaneous

~~Con't):~~

Conclusion:

In each instance, the product of the process are in a more random or disordered state than the reactants

Introduction

In addition to enthalpy, the change in randomness or disorder of a chemical system is extremely important in determining whether equilibrium favours reactants or products.

Introduction Cont.:

The second law of thermodynamics expresses the notion that there is an inherent direction in which any system moves if it is not at equilibrium.

For example

- **If you drop a brick, it falls to the floor**
- **Water placed in a freezer compartment turns into ice**
- **A shiny nail left outdoors eventually rusts**

- Processes that are spontaneous in one direction are not spontaneous in the reverse direction.

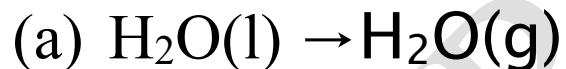
Introduction (cont.):

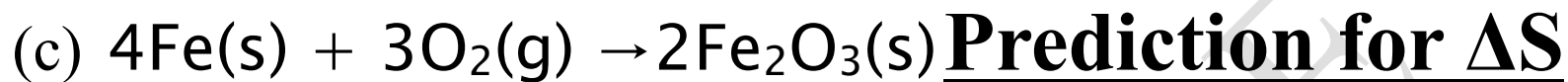
- **SPONTANEITY** is associated with an increase in randomness or disorder of a system.
- This randomness is expressed by a thermodynamic quantity called **ENTROPY**, given a symbol **S** expressed in **units of J/K (not kJ/K)**. The more disordered a system, the larger its entropy.
- **ENTROPY** is an absolute quantity, not a relative quantity. Therefore the entropy of a substance is **S** , not ΔS

• Entropy is a state function. $\Delta S > 0$ indicates an increase in randomness or disorder. $\Delta S < 0$ indicates a decrease in randomness or disorder.

Question

By considering the relative extents of randomness or disorder in the reactants and products, predict whether ΔS is positive or negative for each of the following processes.

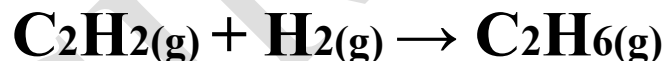




sign:

•It is possible for a system to be disfavoured by enthalpy (endothermic, +ve ΔH), yet be spontaneous because its strongly favoured by entropy (+ve ΔS). For example melting of ice [$\Delta H = +6.01 \text{ kJ/mol}$ and $\Delta S = +22 \text{ J/(K.mol)}$].

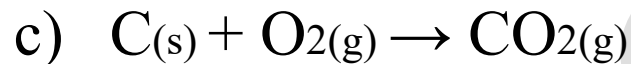
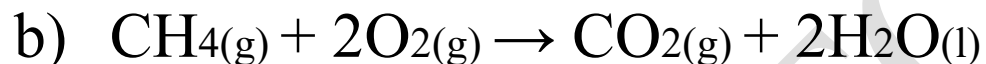
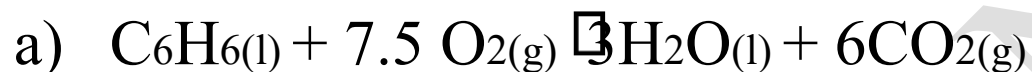
•Ethane, C_2H_6 can be prepared by the reaction of acetylene (C_2H_2) with hydrogen.



Is ΔS° for the reaction positive or negative? Explain.

Prediction for ΔS sign (cont.):

Predict whether ΔS° is likely to be positive or negative for each of the following reactions.



The Second Law of Thermodynamics:

To develop a statement, we must think of an ISOLATED system, one that does not exchange energy or matter with the surroundings.

- **When a process occurs spontaneously in an isolated system, the system always end up in a more random state**

- In the real world, we really deal with isolated systems. We are usually concerned with systems that exchange energy with their surroundings in the form of heat and work. If we consider the entire universe as a giant isolated system, however, we can state the second law as follows:

The Second Law of Thermodynamics:

- **In any spontaneous process, there is always an increase in the entropy of the universe.**
- The second law tells us that the increase in the entropy of the surroundings must be greater than the decrease in the entropy of the system, so that the total entropy increases:

$$\Delta S_{\text{universe}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}} > 0$$

- No process that produce order (a decrease in entropy) in a system can proceed without producing an even larger disorder (an increase in entropy) in its surroundings.

The Second Law of Thermodynamics:

- Thus, while the energy of the Universe is conserved (the first law), the entropy of the Universe continues to increase (the second law)

Other Statements of the Second Law

Clausius statement: It states that heat cannot flow itself from a cold body to a hot body spontaneously without the intervention of an external energy.

Kelvin statement: It is impossible to take heat from a hot body and convert it completely into work by a cyclic process without transferring a part of heat to a cold body.

The Second Law of Thermodynamics:

Thus, the Second Law predicts the direction of heat flow. It explains the reason for why complete conversion of energy into work is not possible.

- It predicts the feasibility of a process
- It introduces two new thermodynamic functions entropy and free energy to explain spontaneity.
- Thus, even if the entropy of a particular system decreases in a spontaneous process, the total change in the entropy (System + Surroundings) must be positive.
- No process that produces order (a decrease in entropy) in a system can proceed without producing an even larger disorder (an increase in entropy) in its surroundings.

The Second Law of Thermodynamics:

- Thus, while the energy of the Universe is conserved (the first law), the entropy of the Universe continues to increase (the second law)
- Molecular systems therefore tend to move spontaneously toward a *state of minimum enthalpy* and *maximum entropy*.

To quantify entropy:

- To calculate disorder or chaotic distribution of energy among molecules requires statistical method – DIFFICULT !!
- However, we can relate the process to the heat involved in a process
- Whenever heat is added to the system, it increases molecular motions causing increased randomness in the system. Thus heat (q) has a randomising influence on the system.
- Can we then equate ΔS with q ? WAIT!
- From experience, distribution of heat also depends on the temperature at which heat is added to the system.

To quantify entropy:

- A system at higher temperature has greater randomness in it than at lower temperature. Thus temperature is the measure of average chaotic motion of particles in the system.
- Heat added to a system at a lower temperature causes greater randomness than when the same quantity is added to it at a higher temperature.
- Thus $\Delta S = \frac{q_{rev}}{T}$
- The change in entropy ΔS of a system is equal to the ratio of heat change q to the temperature (T) of the reversible process.

The Second Law of Thermodynamics:

The change in entropy, dS , in the course of an infinitesimal change is equal to δq_{rev} , the heat absorbed when change is carried out in a reversible manner, divided by the absolute temperature, T as given in the following equation.

$$dS = \frac{\delta q_{rev}}{T}$$

Integration of the above equation yields

$$\Delta S = S_2 - S_1 = \int_{T_1}^{T_2} \frac{\delta q_{rev}}{T}$$



ENTROPY CHANGES FOR AN IDEAL GAS

SIR OTONDE

For an ideal gas

$$\delta q_{\text{rev}} = C_v dT + PdV$$

where only pressure - volume work is involved.

For one mole of gas

$$PV = RT \Rightarrow P = \frac{RT}{V}$$

and thus

$$\delta q_{\text{rev}} = C_v dT + RT \frac{dV}{V}$$

such that

$$dS = \frac{\delta q_{rev}}{T} \text{ yields :}$$

$$\Delta S = S_2 - S_1 = C_v \int_{T_1}^{T_2} \frac{dT}{T} + R \int_{V_1}^{V_2} \frac{dV}{V}$$

$$= C_v \ln \left(\frac{T_2}{T_1} \right) + R \ln \left(\frac{V_2}{V_1} \right)$$

The entropy change for a process involving an ideal gas thus depends on the initial and final temperature and volume.

since

$$\frac{V_2}{V_1} = \frac{T_2 P_1}{T_1 P_2}$$

from the combined gas laws
substitution yields

$$\begin{aligned}\Delta S &= C_v \ln\left(\frac{T_2}{T_1}\right) + R \ln\left(\frac{P_1 T_2}{P_2 T_1}\right) \\ &= C_v \ln\left(\frac{T_2}{T_1}\right) + R \ln\left(\frac{T_2}{T_1}\right) - R \ln\left(\frac{P_2}{P_1}\right)\end{aligned}$$

But

$$\underline{C_v} + R = C_p$$

and thus

$$\Delta S = C_p \ln\left(\frac{T_2}{T_1}\right) - R \ln\left(\frac{P_2}{P_1}\right) \quad \text{for one mole of gas}$$

Consider a case where there is no change in temperature:

$$T_2 = T_1 \quad (\text{i.e. ISOTHERMAL CHANGE})$$

Implies

$$\Delta S_T = R \ln \left(\frac{V_2}{V_1} \right) = -R \ln \left(\frac{P_2}{P_1} \right)$$

For isothermal expansion,

$$V_2 > V_1$$

- Isothermal expansion is accompanied by an increase in entropy. Likewise, isothermal compression is accompanied by a decrease in entropy.

Consider a case where there is no change in pressure:

$$P_1 = P_2 \quad \text{Isobaric process}$$

and

$$\Delta S_p = C_p \ln \left(\frac{T_2}{T_1} \right)$$

Thus, increase in temperature at constant pressure is accompanied by an increase in entropy.

SIR OTONDE

Consider a case where there is no change in volume:

$$V_1 = V_2 \text{ Isochoric change}$$

and one has,

$$\Delta S_V = C_V \ln \left(\frac{T_2}{T_1} \right)$$

Entropy Changes for an Ideal Gas (Examples.) –cont.:

Example:

Two mole of an ideal gas at STP are heated at constant volume to a temperature of 350 K .Determine the increase in entropy for the system. $C_v = 12.47 \text{ J/Mol/K}$)

Solution:

Given volume is constant:

$$\begin{aligned}\Delta S_V &= C_V \ln\left(\frac{T_2}{T_1}\right) \\ \Delta S_V &= 2 \times 12.47 \times \ln\left(\frac{350}{273}\right) \\ &= \underline{\underline{6.20 \text{ J/K}}}\end{aligned}$$

Entropy Changes (Calculations):

Example:

Calculate the entropy change when 1 mole of a perfect gas expands isothermally thus doubling its volume;

Solution:

$$\Delta S = C_p \int_{T_1}^{T_2} \frac{dT}{T} + nR \int_{V_1}^{V_2} \frac{dv}{V}$$

Given that $T_2 = T_1$,

$$\begin{aligned}\Delta S &= nR \ln\left(\frac{V_2}{V_1}\right), (V_2 = 2V_1) \\ &= 1 \times 8.314 \times \ln 2 \\ &= \underline{\underline{5.76 \text{ J/K}}}\end{aligned}$$

Entropy Change accompanying Phase Change:

When a substance changes phase, for example, e.g.
solid $\rightarrow$ liquid or/then liquid $\rightarrow$ vapour;

there is an absorption of heat energy and hence a change in entropy.
Both fusion and vaporization may be carried out reversibly and the heat energy taken up, at constant pressure, is equal to the enthalpy of fusion or vaporization at the given temperature.

For fusion,

$\Delta S_f = \frac{\Delta H_f}{T_f}$; Where ΔH_f = enthalpy of fusion and T_f = freezing temperature.

For vaporisation,

$\Delta S_f = \frac{\Delta H_v}{T_b}$; Where ΔH_f = enthalpy of fusion and T_f = freezing temperature.

where T_b is the boiling point.

Heat is absorbed in both fusion and vaporization and hence both are accompanied by an increase in entropy. The reverse is also true.

Entropy Change accompanying Phase Change (examples):

Example:

Determine the entropy change when one mole of water at boiling point is converted to steam given that the enthalpy of vaporization of water is 40.85 kJ/mol. Solution:

$$\Delta S_v = \Delta H_v / T_b = 40850 / 373$$
$$= 109.5 \quad \text{J/K}$$

Example:

The enthalpy of transition from rhombic to monoclinic sulphur at the transition temperature of 95.6 °C is 0.361 kJ/mole. Determine the entropy of transition. Solution:

$$\Delta S_t = \Delta H_t / T_t = 361 / 368.6 =$$
$$0.98 \text{ J/K}$$

Entropy Change accompanying Phase Change (examples):

Example:

The enthalpy of fusion of monoclinic sulphur is 1.26 kJ/mole. Determine the entropy change when two moles of monoclinic sulphur melts at 392 K.

Solution:

$$\Delta S_f = \frac{\Delta H_f}{T_f} = 2 \times 1260 / 392 = 6.43 \text{ J/K}$$

The Physical Significance of Entropy:

A characteristic of spontaneous processes is that they are accompanied by an increase in the '**disorder**' or 'chaos' of the system. An expansion of a gas into a vacuum is accompanied by an increase of entropy of the system. Since expansion of a gas is accompanied by an increase in disorder, so is the increase in entropy.

Further, **fusion** and **vaporization** are accompanied by increase in entropy. This increase in entropy causes an increase in disorder since the molecules of a liquid are less orderly than those in a solid. Likewise the molecules of vapour are less orderly than those of a liquid. Systems tend to a state of **maximum entropy**. A gas will therefore tend to occupy all the space or volume available to it.

Entropy and the Third Law of Thermodynamics:

Preview of First and second Laws:

First Law:

In any process, spontaneous or nonspontaneous, the total energy of a system and surroundings is constant.

Second Law:

In any spontaneous process, the total energy of a system and surroundings always increases.

While, the third law of thermodynamics concerns the entropy of perfectly-ordered crystals at zero Kelvin.

When a chemical reaction or phase transition is studied at low temperatures, and all substances are pure crystals presumed to be perfectly ordered, the entropy change is found to approach zero as the temperature approaches zero Kelvins.

Mathematical statement of the Nernst heat theorem is expressed as:

$$\lim_{T \rightarrow 0} \Delta S = 0$$

Entropy and the Third Law of Thermodynamics:

SIR OTONDE

Spontaneous processes are accompanied by an increase in the 'disorder' or 'chaos' of the system, or random arrangement of the molecules.

Increase in temperature gives rise to an increase in disorder. Decrease in temperature decreases disorder. Otherwise decrease in temperature results in an increase in order. This leads to the third law of thermodynamics. This law states "***The entropy of a perfect crystalline solid of a pure substance is zero at the absolute zero of temperature***".

The entropy changes of a substance between temperatures T_1 and T_2 are determined from calorimetric measurements then from the definition of entropy increase. $S_{T_2} - S_{T_1} = \int_{T_1}^{T_2} \frac{C_p dT}{T} = C_p \ln \frac{T_2}{T_1}$

This means that the increase in entropy for a substance between 0 K and T is deduced thus:

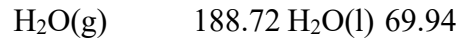
$$S_T = S_0 + \int_0^{T_2} \frac{C_p dT}{T} = C_p \ln T_2$$

Entropy and the Third Law of Thermodynamics (cont.):

The third law then makes it possible to obtain absolute entropies of chemical substances from calorimetric measurements. The standard reaction entropies can then be calculated. The entropy of a reaction is the difference between the entropies of the pure separated products and the pure separated reactants, all substances being in their states at the specified temperature. The entropies of substances are usually reported at 1 atm. pressure and 25°C. These are standard entropies and are indicated by a superscript °.

ENTROPIES OF SOME SUBSTANCES AT 1 ATMOSPHERE AND 25 °C

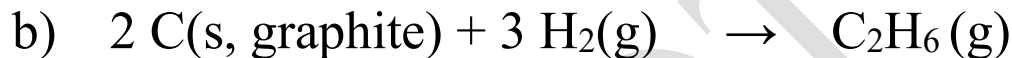
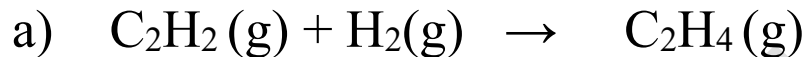
SUBSTANCE	$S^{\circ}(\text{Jmol}^{-1} \text{ K}^{-1})$
H ₂ (g)	130.59
C(c, diamond)	2.44
C(c, graphite)	5.69
C(g)	157.99
CO(g)	197.91
CO ₂ (g)	213.64
CH ₄ (g)	186.19
C ₂ H ₂ (g)	200.82
C ₂ H ₄ (g)	219.45
C ₂ H ₆ (g)	229.49
O ₂ (g)	205.03



Entropy and the Third Law of Thermodynamics (cont.):

Examples

Determine the changes in entropy for the following reactions.



Solutions

a)

$$= 219 - (200.82 + 130.59) =$$

$$\underline{\underline{-112.41\text{J/K}}}$$

b)

$$= 229.3 - (2 \times 5.68 + 3 \times 130.5)$$

$$= \underline{\underline{-173.56 \text{ J/K}}}$$

Summary:

The thermodynamic function, Entropy, is defined as a measure of chaos/disorder. An increase in temperature at constant pressure is accompanied by an increase in entropy. The second law of thermodynamics can be stated as, “heat can not be completely converted into an equivalent amount of work without causing other changes in some part of the system or its surroundings”. The law makes it impossible to ‘freely’ harvest energy from the warm tropical ocean water, for example. The law is also stated as, “All natural or spontaneous processes are thermodynamically irreversible in character”.

If the experimental values of the heat capacities at constant pressure divided by the temperatures (C_p/T) are plotted against the temperatures, then the area under the curve between the initial and the final temperature will give the entropy change.

Alternatively, if the experimental values of the heat capacities at constant pressure (C_p) are plotted against the natural logarithm of the temperatures, then the area under the curve between the logarithms of the initial and final temperature gives the entropy change.

The entropy change for a process involving an ideal gas depends on the initial and final temperature and volume.

The efficiency of the engine is the fraction of the heat taken in at the higher temperature that is converted into work.

SAQs

1. 10 g of ice at 273K is added to 20g of water at 363K in an insulated flask. The heat of fusion of ice is 6000 Joule per mole and the specific heat capacity of water is 4.2 joule per Kelvin per gram. Ignoring the heat capacity of the flask: (a). Determine the final temperature of the system (b). Determine ΔS for the system

2. Calculate the entropy change for the reaction:

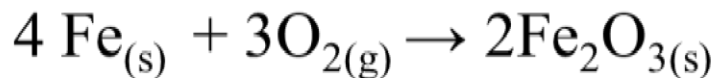
$2\text{C}_{(\text{graphite})} + 2\text{H}_{2(\text{g})} = 2\text{C}_2\text{H}_{4(\text{g})}$ given the following standard entropies at 25 °C in units $\text{JK}^{-1}\text{mole}^{-1}$;

$\text{C}_{(\text{graphite})}$	5.7
$\text{H}_{2(\text{g})}$	131.2
	221.0

$\text{C}_2\text{H}_{4(\text{g})}$ Answers:

[1] (a) 306.5 K. [2] (b) 2.72 JK^{-1} .
 -52.8 JK^{-1} .

Determine whether the reaction is spontaneous: Consider the oxidation of iron according to the equation below:



By determining the sign of ΔS_{total} , show whether the reaction is spontaneous at 25 °C.

Solution:

To determine the sign of $\Delta S_{\text{total}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}}$, the sum values are calculate as follows:

$$\begin{aligned}\Delta S_{\text{sys}} &= \Delta S^\circ = 2 S^\circ[\text{Fe}_2\text{O}_3] - [4S^\circ(\text{Fe}) + 3 S^\circ(\text{O}_2)] \\ &= 2 S^\circ[87.4] - [4(27.3) + 3(205)] = -549.5 \text{ J/K}\end{aligned}$$

$$\Delta H^\circ = 2 \Delta H^\circ_f(\text{Fe}_2\text{O}_3) = 2 (-824.2) = -1648.4 \text{ kJ}$$

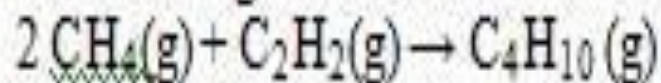
$$\text{Thus, } \Delta S_{\text{surr}} = \frac{-(-1648400)}{298.15} = 5529 \text{ J/K}$$

Therefore,

$$\Delta S_{\text{total}} = -549.5 \text{ J/K} + 5529 \text{ J/K} = 4980 \text{ J/K}$$

The reaction is spontaneous under standard conditions since ΔS_{total} is +ve.

- (a) (i) Calculate the standard entropy change at 298 K of the system and of the surroundings for the reaction



Given the following data:

ΔH_f° (kJ/mol) S° (J/mol-K)

$\text{CH}_4(\text{g})$	-74.8	186.3
$\text{C}_2\text{H}_2(\text{g})$	226.7	200.8
$\text{C}_4\text{H}_{10}(\text{g})$	-124.73	310.0

[6 marks]

- and (ii) The total entropy change for the system and surrounding.

[2 marks]

- (iii) Is the production of butane gas (C_4H_{10}) at 298 K from the reaction of methane (CH_4) with ethyne (C_2H_2) a spontaneous process? Explain.

[2 marks]

Entropy : Examples:

**CONSIDER A SYSTEM HELD AT CONSTANT T AND
CONSTANT P CAPABLE OF PV WORK**

$$\Delta S \geq \frac{q_p}{T}$$

$$0 \geq q_p - T\Delta S$$

$$0 \geq \Delta E - T\Delta S$$

$$\text{or } q_p - T\Delta S \leq 0$$

$$\Delta H - T\Delta S \leq 0$$

$$\equiv H - T.S$$

and $H = E + PV$ In
a similar manner:

$$G_1 = H_1 - T S_1$$

$$G_2 = H_2 - T S_2$$

$$\Delta G = \Delta H - T \Delta S$$

For an isobaric change, the pressure is constant and we
have: $\Delta H = \Delta E + P \Delta V$

$$\text{and hence } \Delta G = \Delta H + P \Delta V - T \Delta S$$

$$\Delta G = \Delta E - T \Delta S + P \Delta V$$

Implying that

$$\Delta G = \Delta A + P \Delta V$$

$$-\Delta G = w_{rev} - P \Delta V$$

$$-\Delta G = w_{net}$$

• **Thermodynamics tells us that the *change in free energy for a process, ΔG , equal the maximum useful work that can be done by the system on its surroundings in a spontaneous process occurring at constant temperature and pressure.***

• $-\Delta G = w_{net}$ (net work obtainable (at constant T and P) for the system.

- The function, **G**, is known as the **Free Energy**.

VARIATION OF FREE ENERGY WITH TEMPERATURE OR PRESSURE

We have already seen that

$$H = E + PV$$

$$G = H - TS$$

Therefore one has

$$G = E + PV - TS$$

$$\begin{aligned} \text{For small changes, } dG &= dE + PdV \\ &+ VdP - TdS - SdT \end{aligned}$$

From the 1st Law, $\delta q = dE + w$

If

w is due to volume change, then

$$\delta q = dE + PdV$$

If it is assumed that the process is reversible, P is the pressure of the system and further $q/T = dS$

meaning

$$TdS = \delta q$$

Such that

$$TdS = dE + PdV$$

Then

$$dG = dE + PdV + VdP - TdS - SdT$$

$$dG = TdS + VdP - TdS - SdT$$

$dG = VdP - SdT$ at constant temperature, $dT = 0$

$$V = \left(\frac{\partial G}{\partial P} \right)_T$$

For an appreciable process taking place at constant temperature, between two pressures P_1 and P_2 , we can obtain the free energy change by integrating:

$$\int_{G_1}^{G_2} dG = \int_{P_1}^{P_2} VdP$$

Since temperature is constant and

$$PV = RT \quad V =$$

RT/P then

$$\Delta G = RT \int_{P_1}^{P_2} \frac{dP}{P}$$

$$= RT \ln \frac{P_2}{P_1}$$

For n moles ,

$$\Delta G = nRT \ln \left(\frac{P_2}{P_1} \right)$$

Example

Calculate the change in free energy when 11.21 dm³ of a perfect gas at 0°C and 760 mmHg pressure expands isothermally until its pressure is 190 mmHg.

Solution

$$T = 273 \text{ K}, R = 8.314 \text{ J/mol/K}, P_1 = 760 \text{ mm Hg}, P_2 = 190 \text{ mm Hg}$$

$$n = PV/RT = (101325 \times 11.21 \times 10^{-3})/(8.314 \times 273) \text{ n}$$

$$= 0.5 \text{ mol}$$

Then

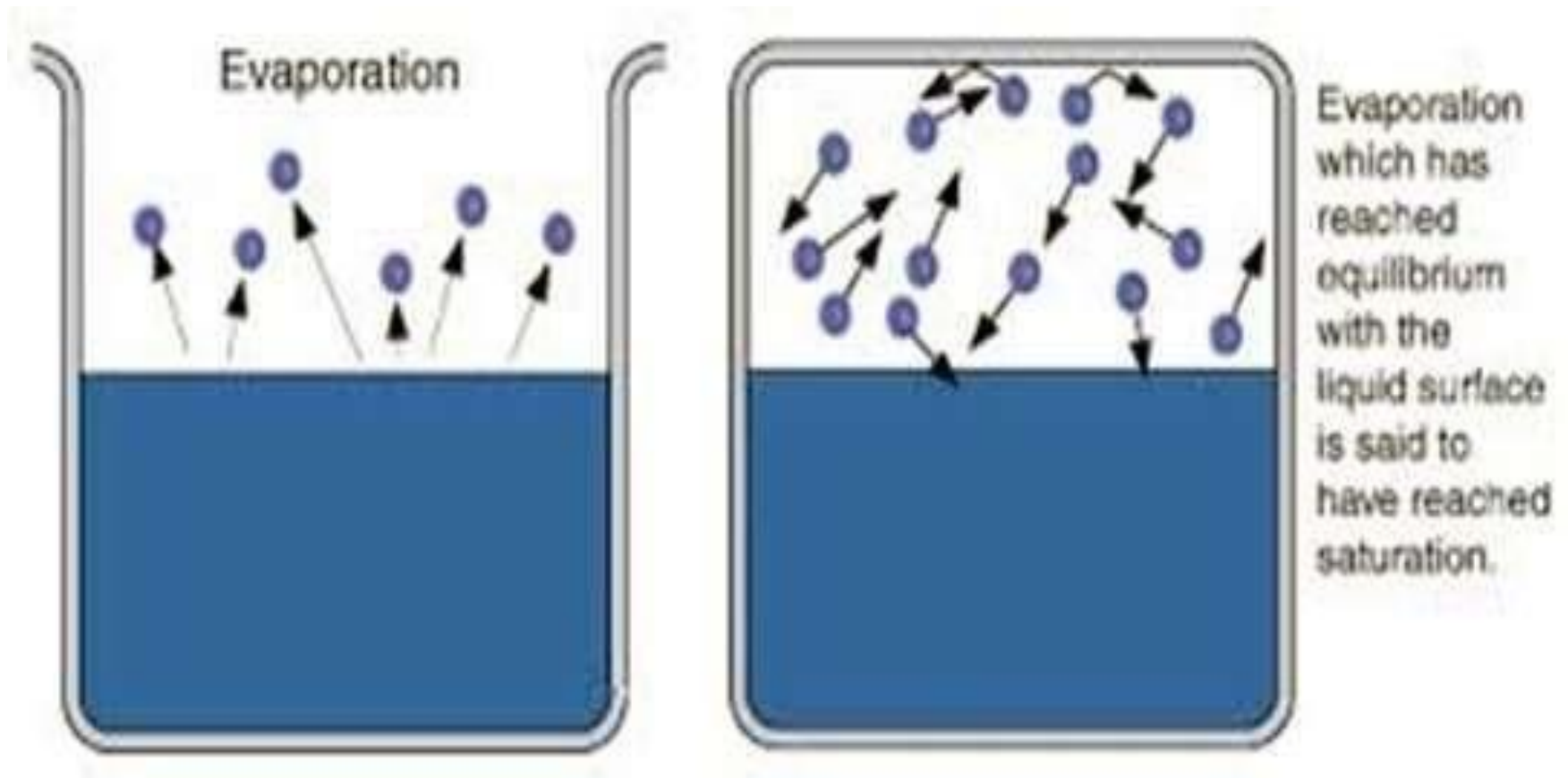
$$\Delta G = nRT \ln (P_1/P_2)$$

$$= 0.5 \times 8.314 \times 273 \ln(190/760)$$

$$= -1573.3 \text{ J Ans.}$$

PHASES IN EQUILIBRIUM: THE CLAUSIUS - CLAPEYRON EQUATION

Consider two phases of the same substance in equilibrium with one another at temperature T and Pressure P



Consider a liquid in equilibrium with its vapors at a temperature T and pressure P .

At equilibrium $G_A = G_B$ where A and B signify the two phases. For a small change in P and T Thus:

$$T \rightarrow T + dT \quad P \rightarrow P + dP.$$

For an infinitesimal change where the only work done is that due to expansion, one has: $dG_A = V_A dP - S_A dT$

$$dG_B = V_B dP - S_B dT$$

•One can see that, at equilibrium

$$V_A dP - S_A dT = V_B dP - S_B dT$$

and hence

$$(V_B - V_A) dP = (S_B - S_A) dT$$

$$\Rightarrow \frac{dP}{dT} = \frac{S_B - S_A}{V_B - V_A} = \frac{\Delta S}{\Delta V}$$

$S_B - S_A$ is the entropy change accompanying phase change. If B is the phase which tends to be formed as T is raised, then as stated

earlier,

per

$$\Delta S = \frac{\Delta H}{T} = \frac{dq}{T}$$

$$\therefore S_B - S_A = \frac{\Delta H}{T}$$

□

where H is the enthalpy absorbed mole in the phase change. This implies that:

$$\frac{dP}{dT} = \frac{\Delta H}{T(V_B - V_A)}$$

This is the general form of the **Clapeyron - Clausius** Equation.

For conversion of liquid to vapour

$$\Delta H = H_V \text{ and } V_B \gg V_A$$

$$\square V_B - V_A \square V_B = V$$

For a mole of water for example,

$V_{\text{vapour}} = 22400 \text{ ml}$ while $V_{\text{liquid}} = 18 \text{ ml}$

Such that $V_{\text{vapour}} - V_{\text{liquid}} = 22382 \text{ ml}$ $\square V_{\text{vapour}}$

Hence

$$\frac{dP}{dT} = \frac{\Delta H_v}{TV}$$

$\Delta H_v \equiv$ molar enthalpy change of vaporization.

.

•For a mole of gas

$$PV = RT$$

$$\Rightarrow V = \frac{RT}{P}$$

Rearrangement and integration gives:

$$\frac{dP}{dT} = \frac{\Delta H_v \cdot P}{T^2 R}$$

$$\int_{P_1}^P \frac{dP}{P} = \frac{\Delta H_v}{R} \int_{T_1}^{T_2} T^{-2} dT$$

$$\ln\left(\frac{P_2}{P_1}\right) = \frac{\Delta H_v}{R} \left(\frac{1}{T_1} - \frac{1}{T_2}\right) = \frac{\Delta H_v}{R} \left(\frac{T_2 - T_1}{T_2 T_1}\right)$$

Further

$$\frac{dP}{dT} = \frac{\Delta H_v P}{RT^2}$$

$$\Rightarrow \int \frac{dP}{P} = \frac{\Delta H_v}{R} \int T^{-2} dT$$

$$\Rightarrow \ln P = \frac{-\Delta H_v}{RT} + C$$

Thus a plot of the natural logarithm of the vapour pressure against the reciprocal of temperature should



give a straight line graph with slope equal to $(-H_v/R)$ from which the molar enthalpy of vaporization (H_v) may be determined.

STANDARD FREE- ENERGY CHANGES (ΔG°)

Free energy is a state function. It is possible to tabulate **standard free energies of formation** for substances.

Temperature chosen for tabulating data is 25°C but ΔG° at other temperatures may be calculated

The standard free energies of formation for substances are useful in calculating the standard free-energy in a chemical process.

Thus $\Delta G^\circ = \sum nG^\circ_f(\text{products}) - \sum mG^\circ_f(\text{reactants})$

STANDARD FREE ENERGIES

STATE OF MATTER.	STANDARD STATE
Solid	Pure solid
Liquid	Pure liquid
Gas	1 atm pressure

Solution	1 M concentration
Elements	Standard free energy of formation of an element in its normal state is defined as zero

STANDARD FREE ENERGIES

EXAMPLE

Determine the standard free-energy change for the following reaction at 298 K: $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$

SOLUTION:

From tables, for $\text{N}_2(\text{g})$, $\Delta G^\circ_f = 0.0$; $\text{H}_2(\text{g})$, $\Delta G^\circ_f = 0.0$; $\text{NH}_3(\text{g})$, $\Delta G^\circ_f = -16.66 \text{ kJ/mol}$

$$\Delta G^\circ = 2\Delta G^\circ_f(\text{NH}_3) - [3\Delta G^\circ_f(\text{H}_2) + \Delta G^\circ_f(\text{N}_2)]$$

Inserting numerical quantities, we obtain $\Delta G^\circ = -33.32 \text{ kJ}$

TABLE OF THE STANDARD MOLAR FREE ENERGIES OF FORMATION OF SOME COMPOUNDS AT 298K

SUBSTANCE	$\Delta G^\circ_{\text{of}(298)} , \text{ kJ}$
$\text{H}_2\text{O}(\text{l})$	- 236.9650
$\text{CH}_3\text{OH}(\text{l})$	- 166.0714
$\text{C}_2\text{H}_5\text{OH}(\text{l})$	- 174.5986

$\text{C}_6\text{H}_6(\text{l})$	+ 124.3801
$\text{NaCl}(\text{s})$	- 383.6613
$\text{H}_2\text{O}(\text{g})$	- 228.3772
$\text{C}_2\text{H}_2(\text{g})$	+ 209.0
$\text{C}_2\text{H}_6(\text{g})$	- 32.8548
$\text{CO}_2(\text{g})$	- 394.0040
$\text{CO}(\text{g})$	- 137.1370

Standard Free-Energy Changes for Reactions

Calculate the standard free-energy change at 25 °C for the Haber synthesis of ammonia using the given values for the standard enthalpy and standard entropy changes:



$$\Delta S^\circ = -198.7 \text{ J/K}$$

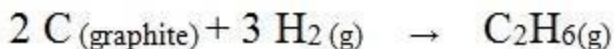
$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$= -92.2 \text{ kJ} - \left(298 \text{ K} \times \frac{-198.7 \text{ J}}{\text{K}} \times \frac{1 \text{ kJ}}{1000 \text{ J}} \right) = \boxed{-33.0 \text{ kJ}}$$

You are given the following entropies and enthalpies of combustion at 25°C;

Substance	S (JK ⁻¹)	ΔH (kJ)
C _(graphite)	5.9	-396
H _{2(g)}	131.0	-287
C ₂ H _{6(g)}	231.0	-1567

State whether the following reaction is thermodynamically possible.



Solution (Students to work this out!)

Step I: Determine the entropy change for the reaction.

Step II: Determine the enthalpy change for the reaction **Step III.**

Determine the Gibb's free energy change for the reaction.

Final Step: From the value of Gibb's free energy change comment on the spontaneity of the reaction, and hence give the answer to the question.

•**SUMMARY**

- Gibb's free energy change (ΔG) is a better determinant of spontaneity than entropy and enthalpy change.
- The condition for spontaneity is that ΔG must be negative.
- At equilibrium, ΔG equals zero.
- Standard Gibb's Free Energy, G , is taken as zero for all elements in their most stable form at a temperature of 298K and a pressure of one atmosphere.
- The Clausius-Clapeyron equation can be used to predict the equilibrium vapour pressure P_2 of a pure liquid at a desired

temperature T_2 from its known enthalpy of vaporization and its known vapour pressure P_1 at some other temperature T_1 .

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